

AEBN21:

Public Buildings -
Integrating Solar Energy, Costs and Environmental Issues

Department of Architecture and Built Environment, Energy and Building Design

Department of Building and Environmental Technology, Building Services

Autumn 2025 | MEEBD

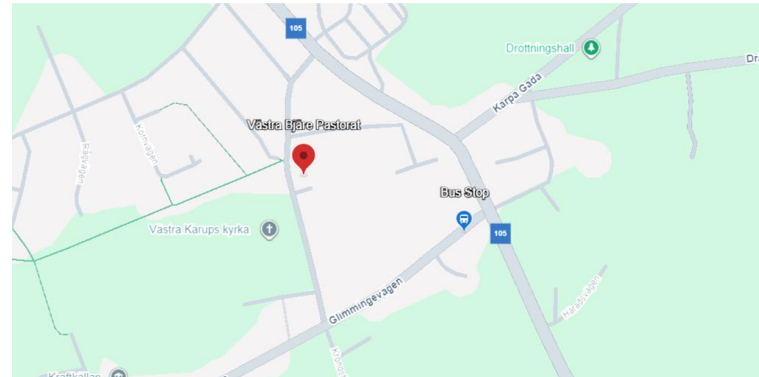
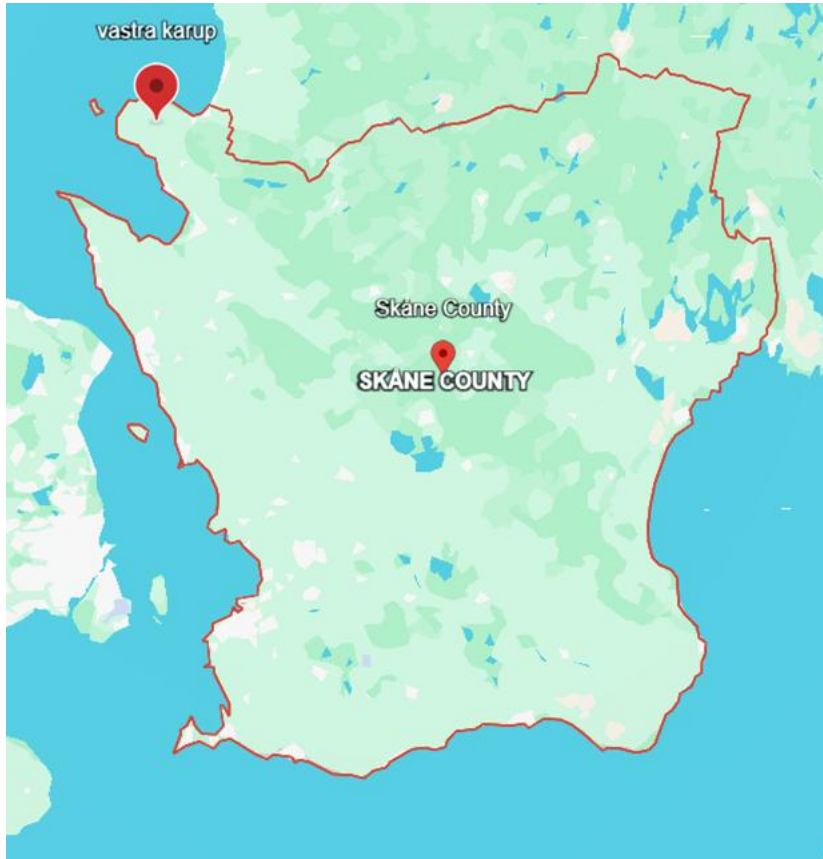
Date: 17th December, 2025

Group No. 10

Himadri, Kamal, Vinay



SITE INTRODUCTION



Location:

56°24'41"N 12°44'33"E

Församlingsvägen 9, Västra Karup,
Skane, Sweden.

About:

Known as Västra Bjäre Pastorat,
houses the parish office and a
community center equipped with
kitchen.



RESEARCH QUESTION



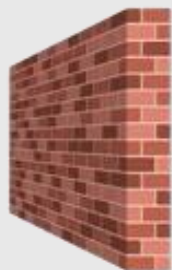
- What strategies best save money and energy? Do they also help the environment?
- Is there a practical middle ground?
- Does the best strategy change with building function?



STRATEGIES

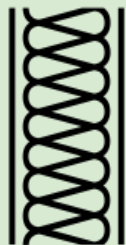


BASE CASE

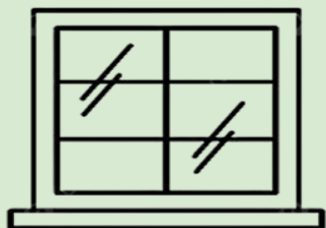


1

PASSIVE



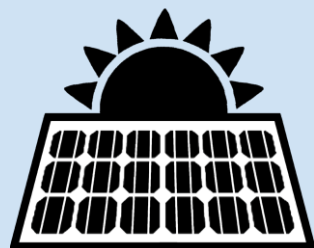
2 x 3 x 2



2

54 Parameters

ACTIVE



3

162 Parameters

BEHAVIOURAL
OCCUPANCY BASED ON FUNCTION

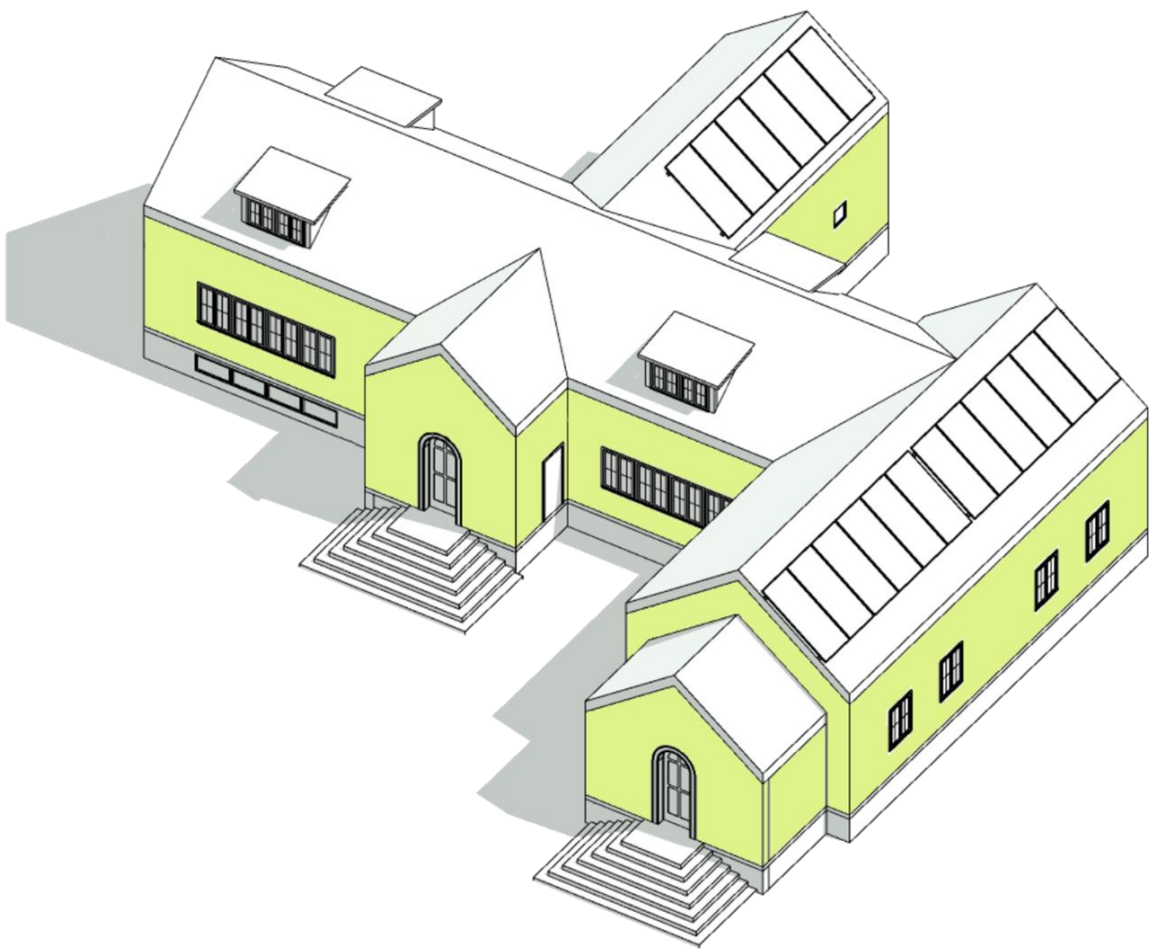


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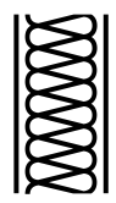
324 Parameters



PASSIVE STRATEGIES



INSULATION



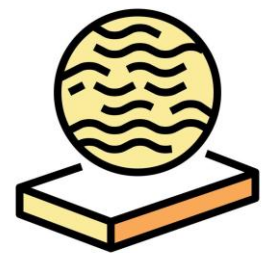
Types



Location



Thickness(in mm)



Glass mineral wool



Hemp insulation



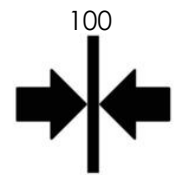
EXTERNAL WALL INSULATION

External

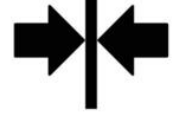


THERMAL INSULATION

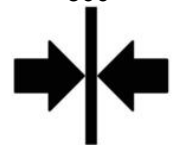
Internal



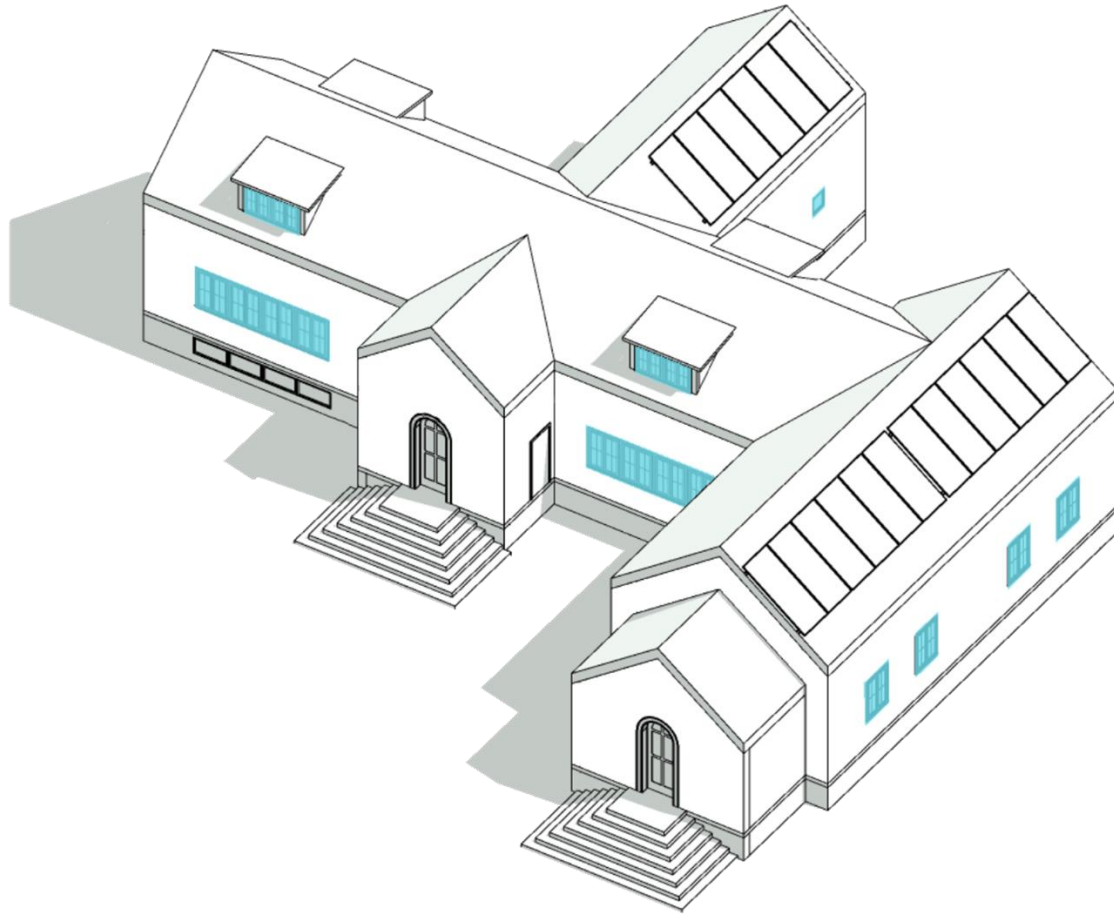
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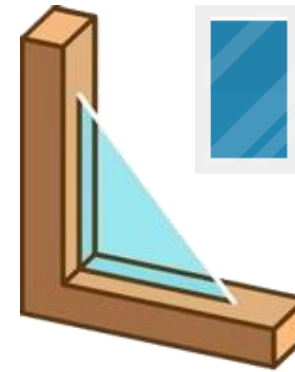
300



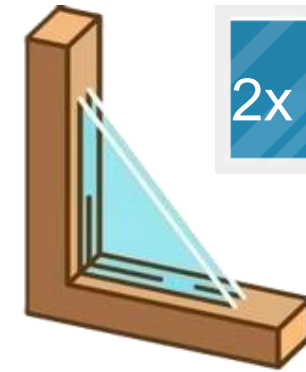
PASSIVE STRATEGIES



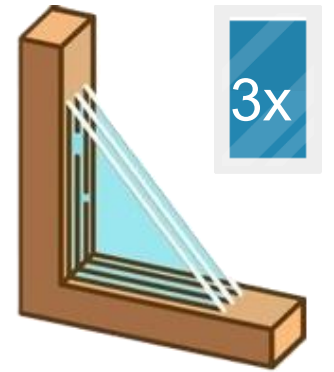
WINDOWS



Existing
Considered
double glazed



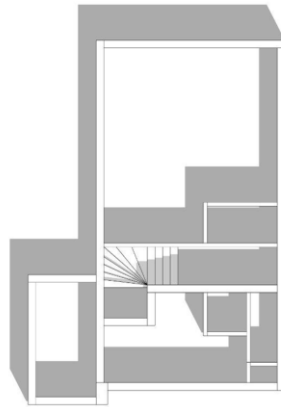
Double Glazing
windows
produced with
aluminium
profiles
manufactured
by AEA
members.



Triple Glazing
Top-swing window
3-glass Alu From
Elitfönster AB



PASSIVE STRATEGIES- Considerations

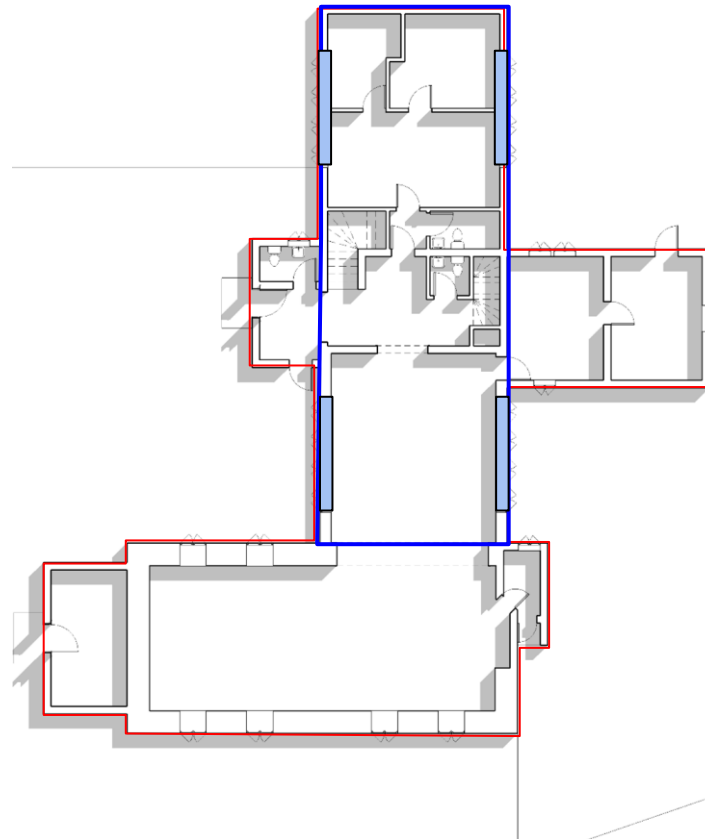


Basement floor

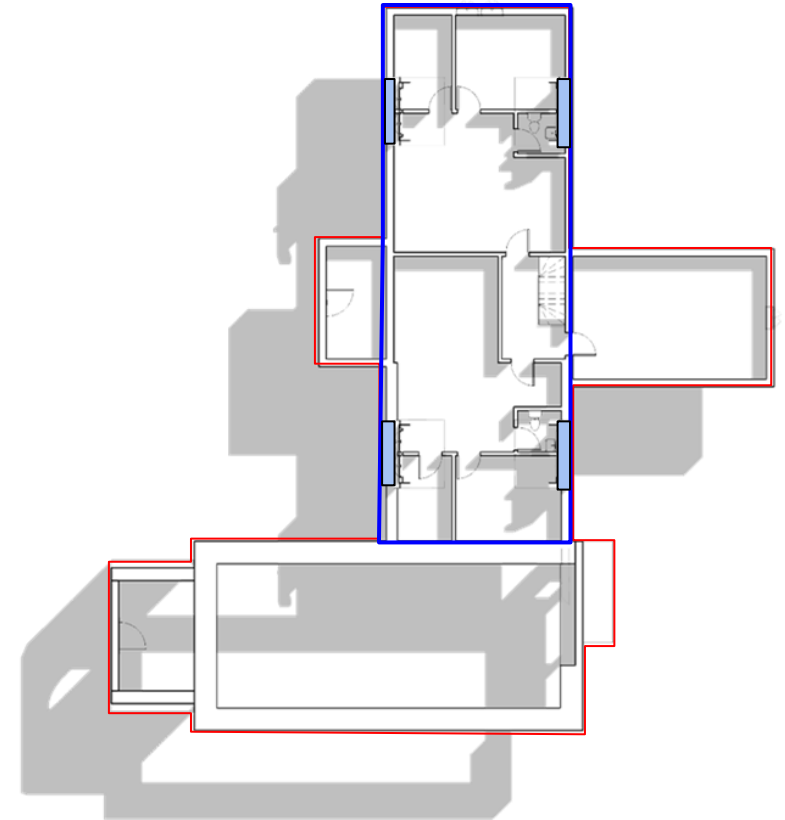
- Entire facade-438 m²
- Considered facade 193 m²

Insulation and windows considered for
the apartment areas only

- All facade
- Considered windows



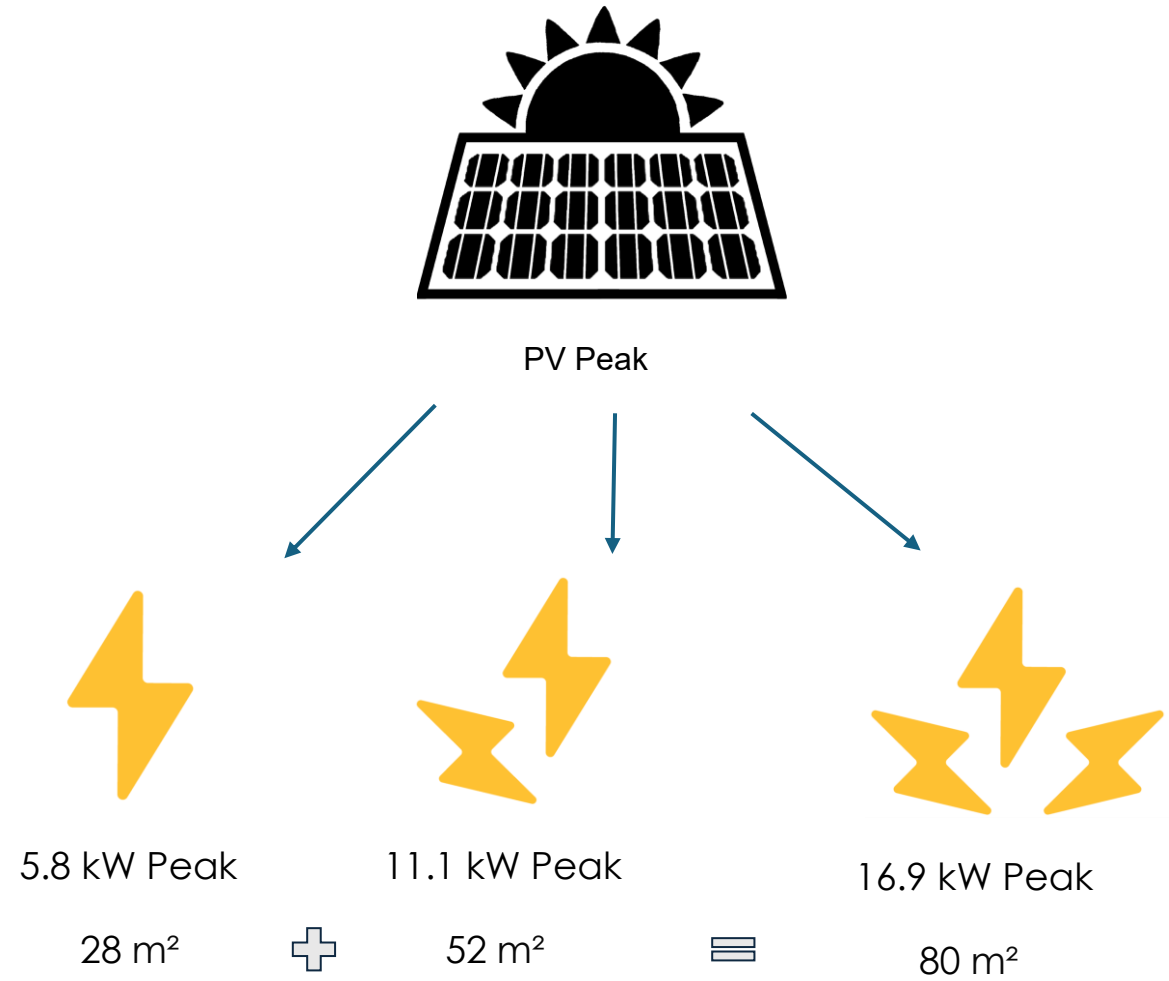
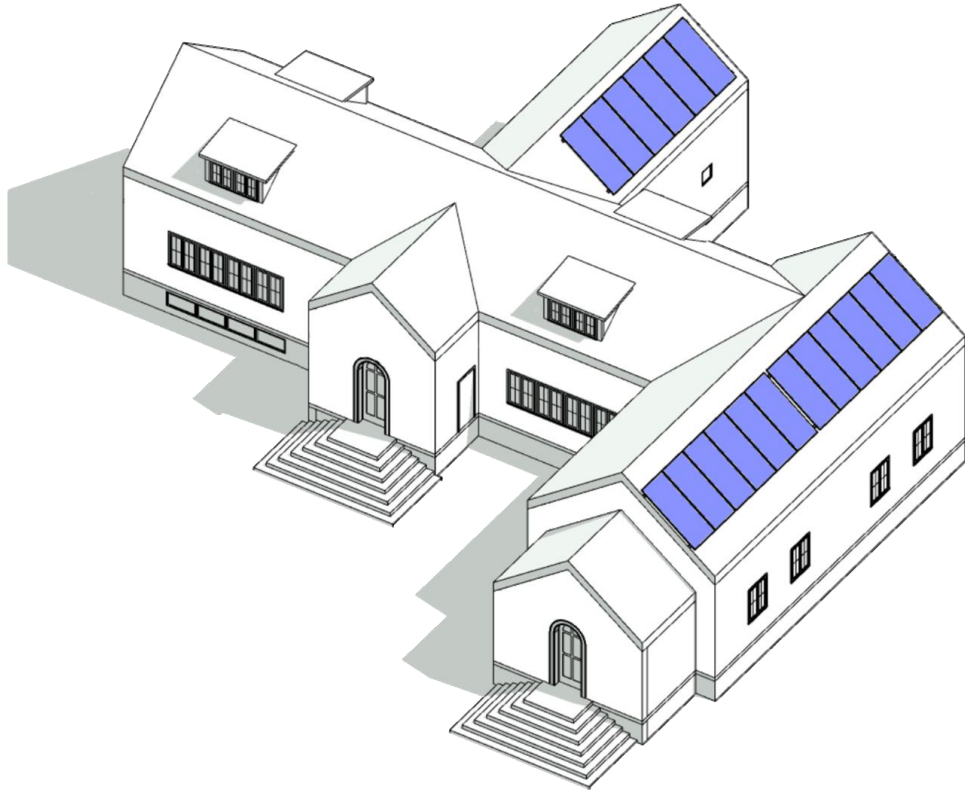
First floor
Considered facade area - 151 m²



Second floor
Considered facade area - 42 m²



ACTIVE STRATEGIES



BEHAVIOURAL STRATEGIES

Input Parameters

People Density

Occupancy Schedule

ACH

Lighting

Equipment load

FUNCTION



THERMAL ZONES



UNOCCUPIED

- ENTRY PORCH
- FOYERS
- BATHROOMS
- KITCHEN
- PUMP ROOM
- CORRIDORS
- ATTIC



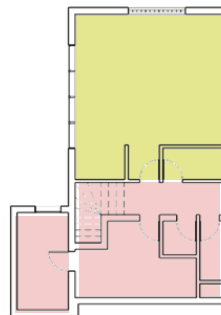
SEMI- OCCUPIED

- HOBBY ROOM

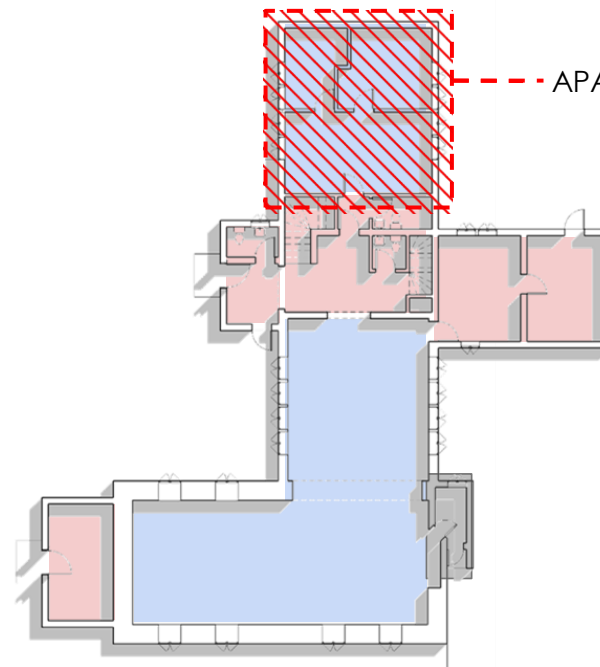


OCCUPIED

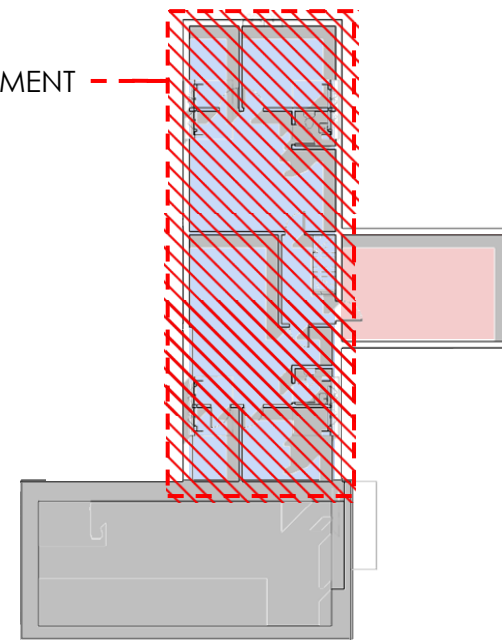
- MAIN HALL
- DINING HALL
- OFFICES ON FF
- OFFICES ON SF



Basement



First floor

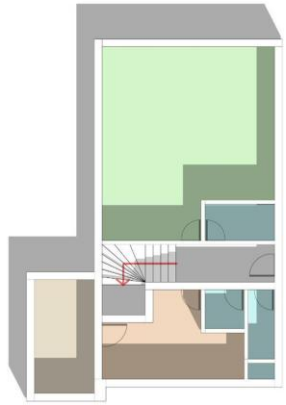


Second floor

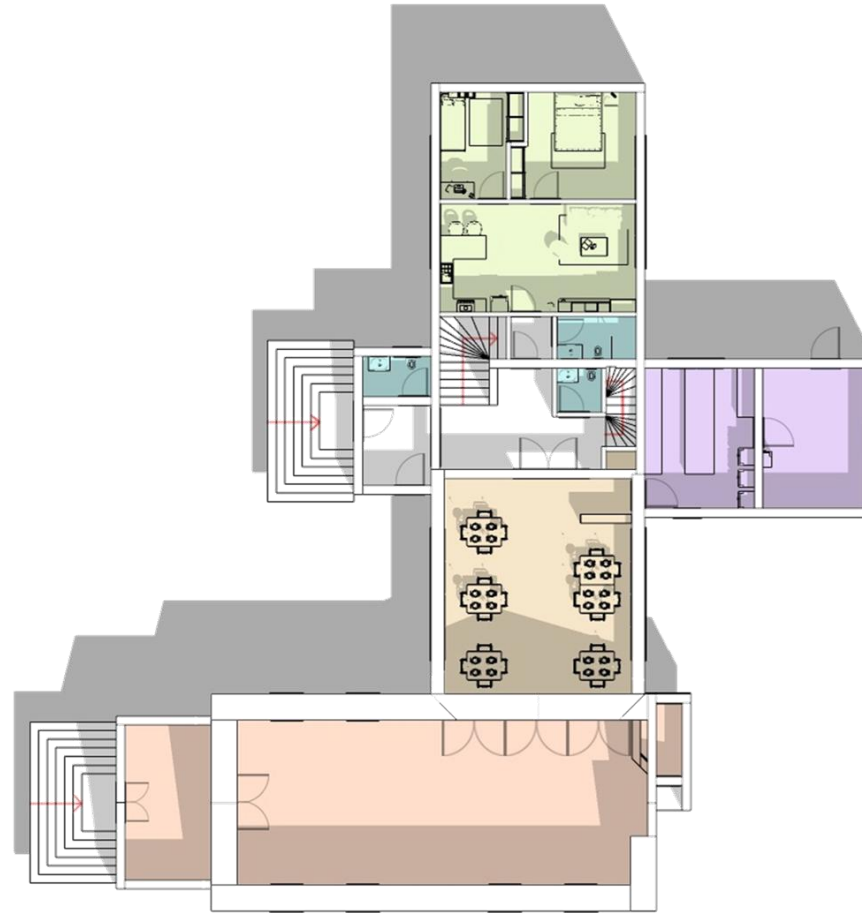
Total of office/apartment area - 208.5 m



PROPOSED LAYOUT

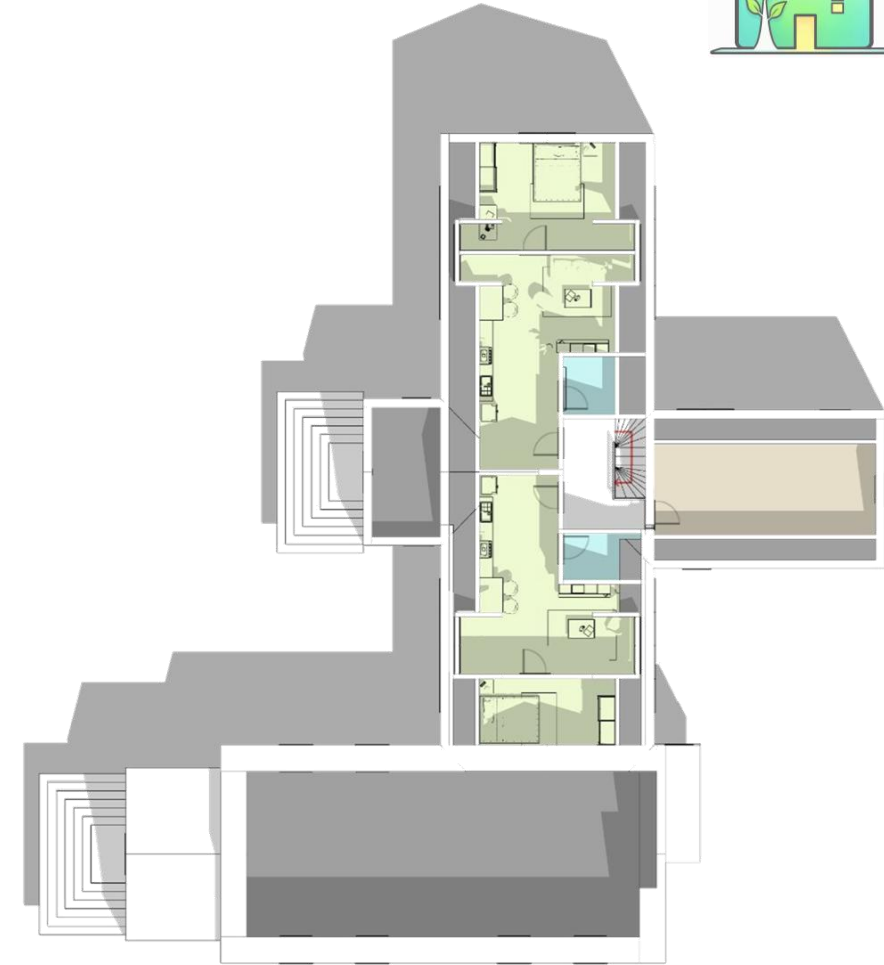


Basement



First Floor

51.5 m² Apartment area



Second Floor

157 m² Apartment area
Effective area - 80 m²



PERFORMANCE CRITERIA



BUILDING ENERGY (In kWh/m²·year)



Inclusive of Heating , lighting, equip and PV

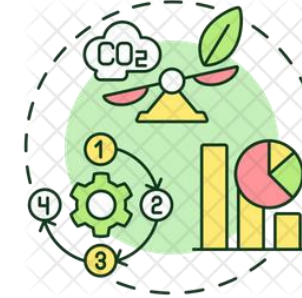
LIFE CYCLE COST (In SEK)



Assumptions:
N - 50 year life cycle

Rdi - 3.14%

LIFE CYCLE ASSESSMENT (in kg CO₂ eq.)



Global Warming
Potential

Assumptions:
50 year life cycle

Functional units same for window and insulation resp.

Environmental impact - Global warming potential

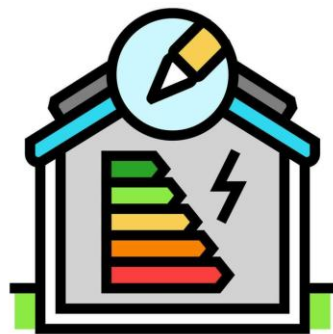
Cradle to grave(without D)



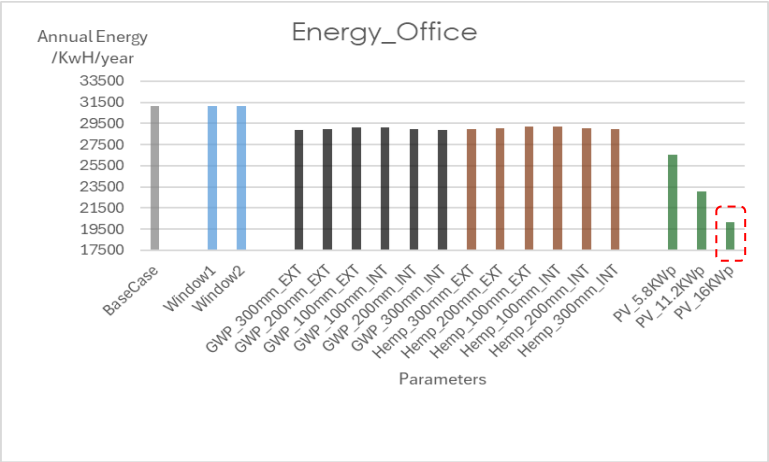
RESULTS



BUILDING ENERGY (In kWh/m²·year)



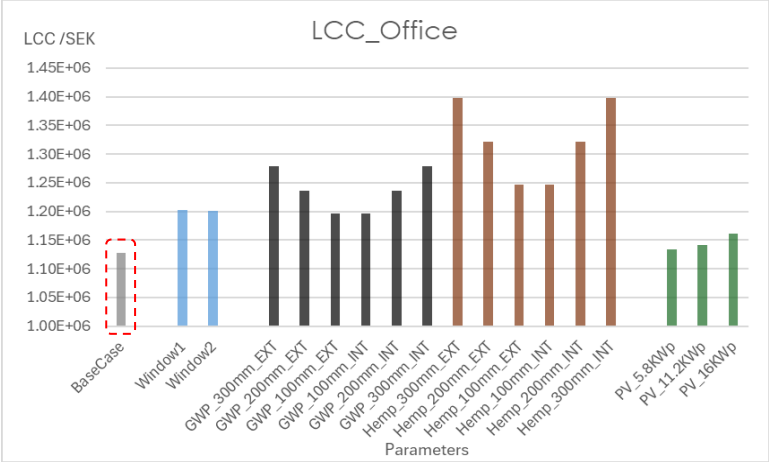
16.9 KW Peak PV
With lesser than 20k kWh/year
energy



LIFE CYCLE COST (In SEK)



BASE CASE
with around 1.15 Million SEK

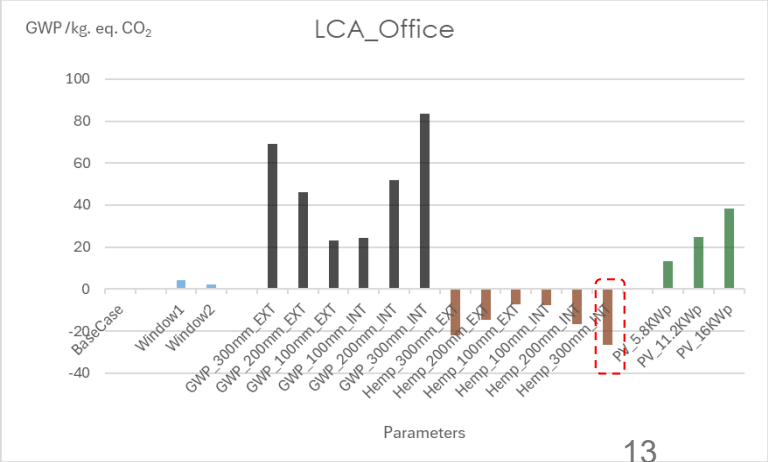


LIFE CYCLE ASSESSMENT (in kg CO₂ eq.)

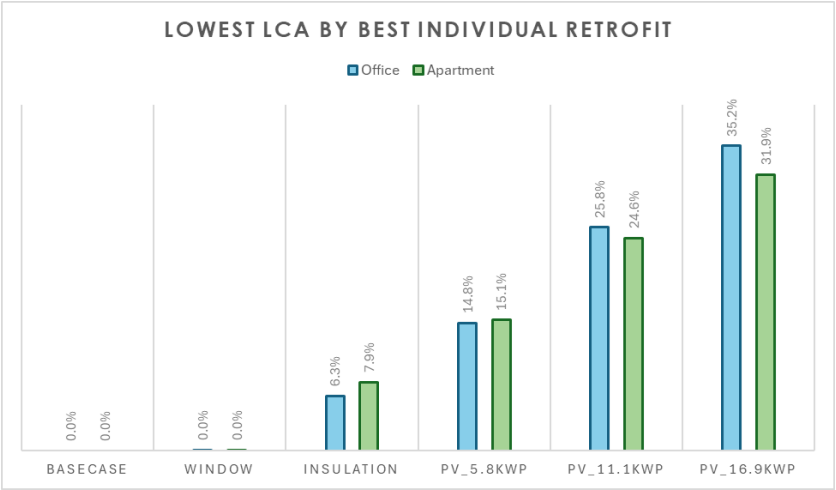
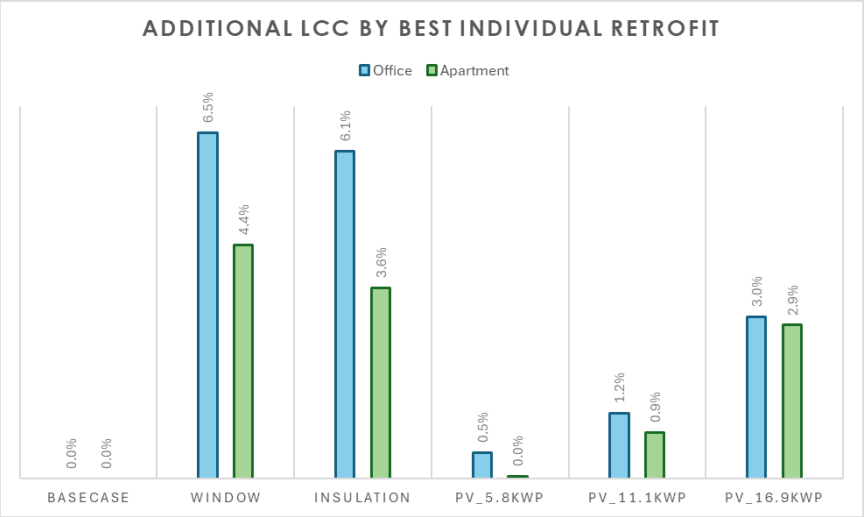
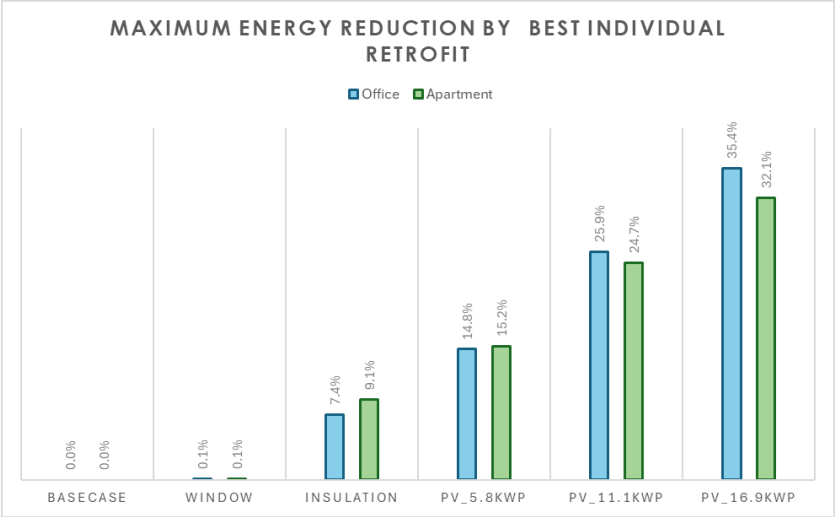


Global Warming
Potential

300mm thick Hemp Insulation
placed in the interior side.
-26.6 kg CO₂ eq.



RESULTS

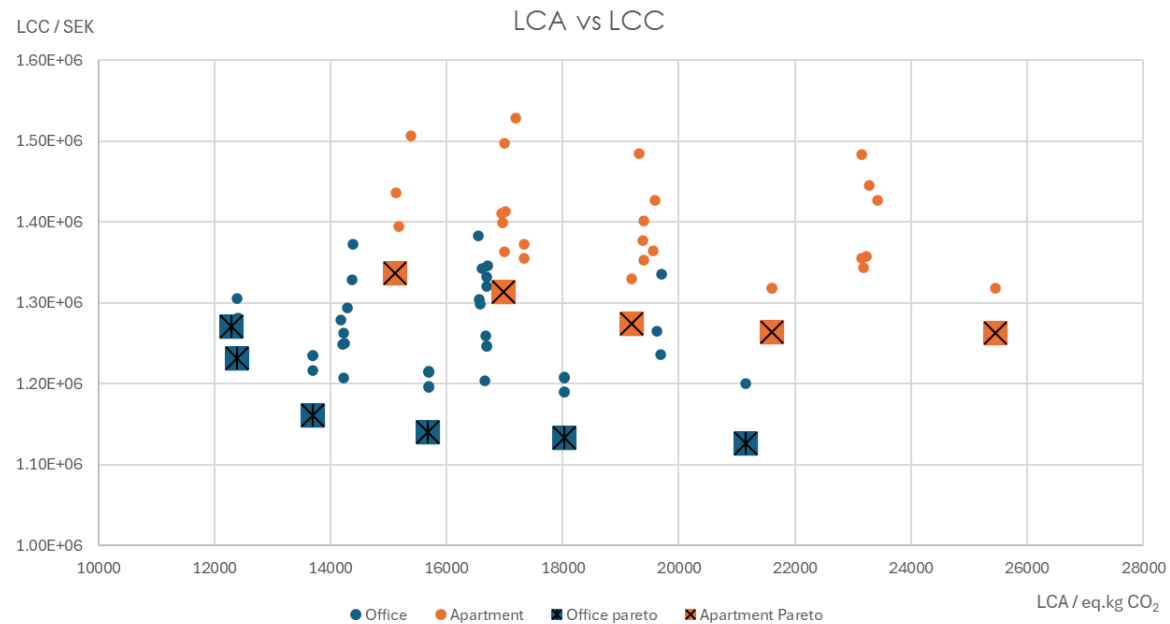
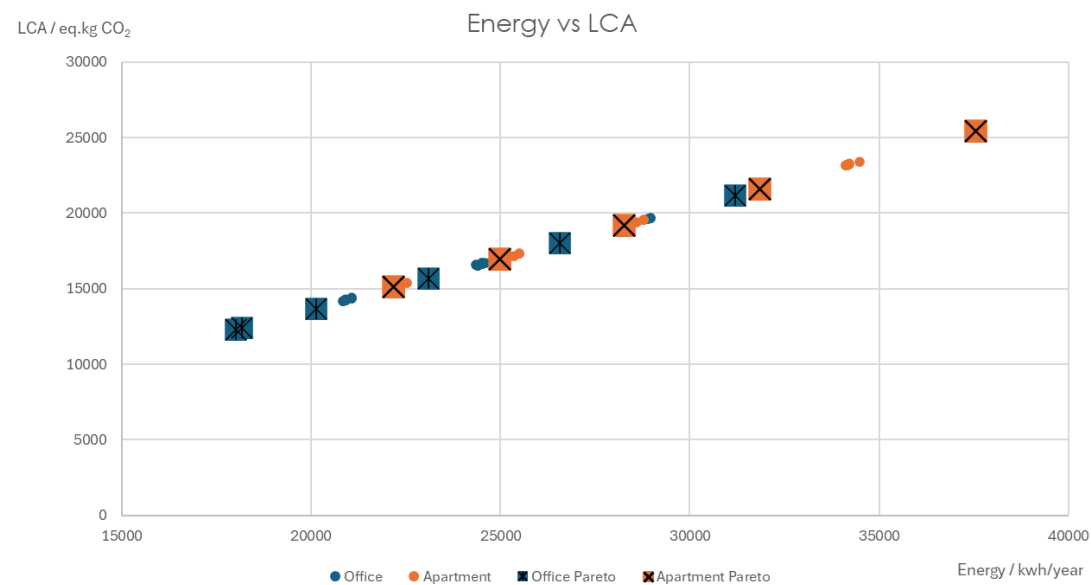
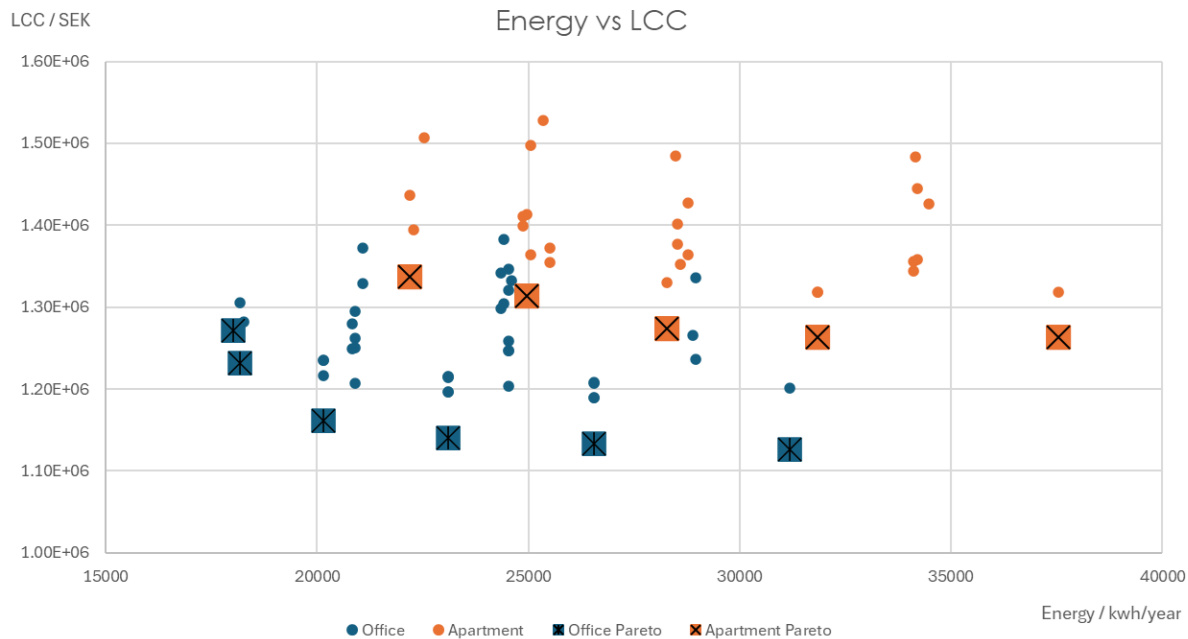


Office



Apartment





Office



Office Pareto



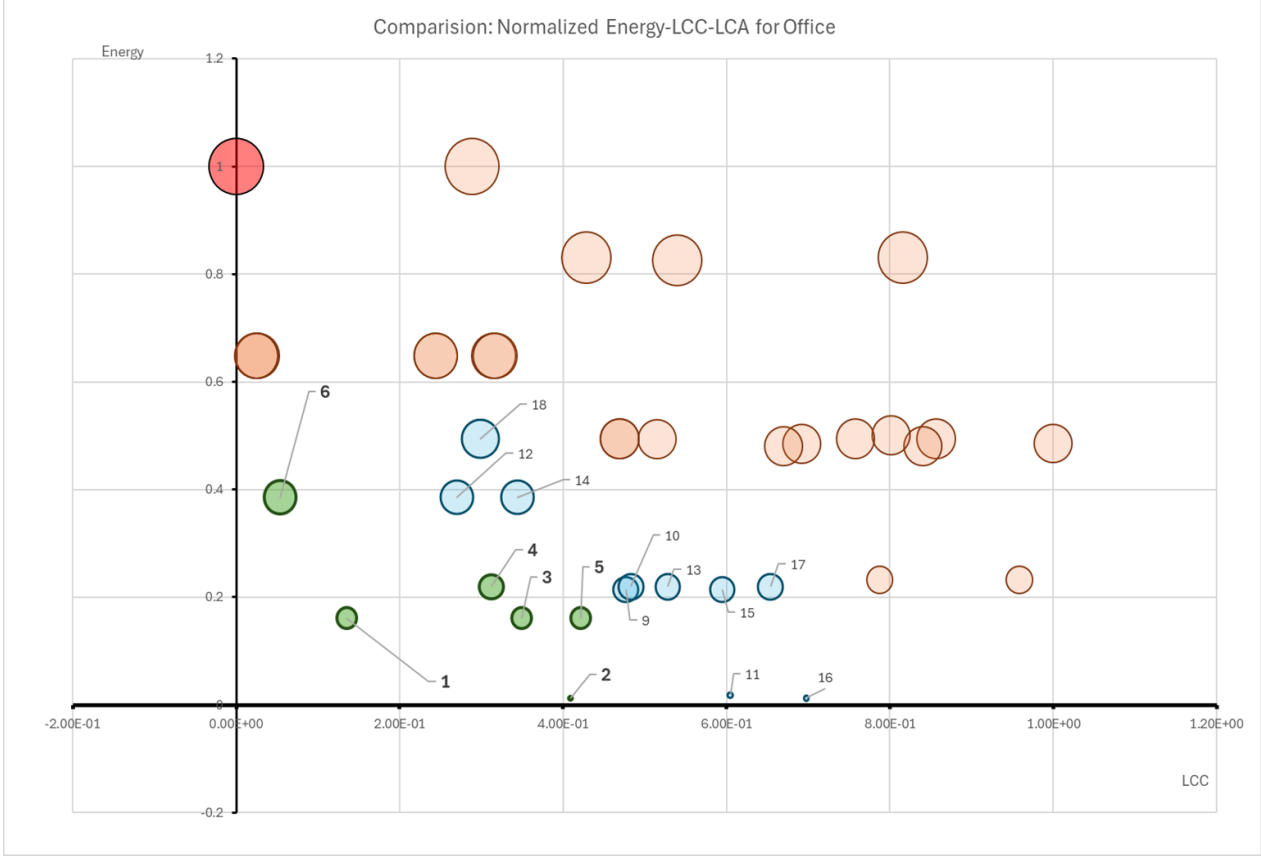
Apartment



Apartment Pareto



RESULTS - Office



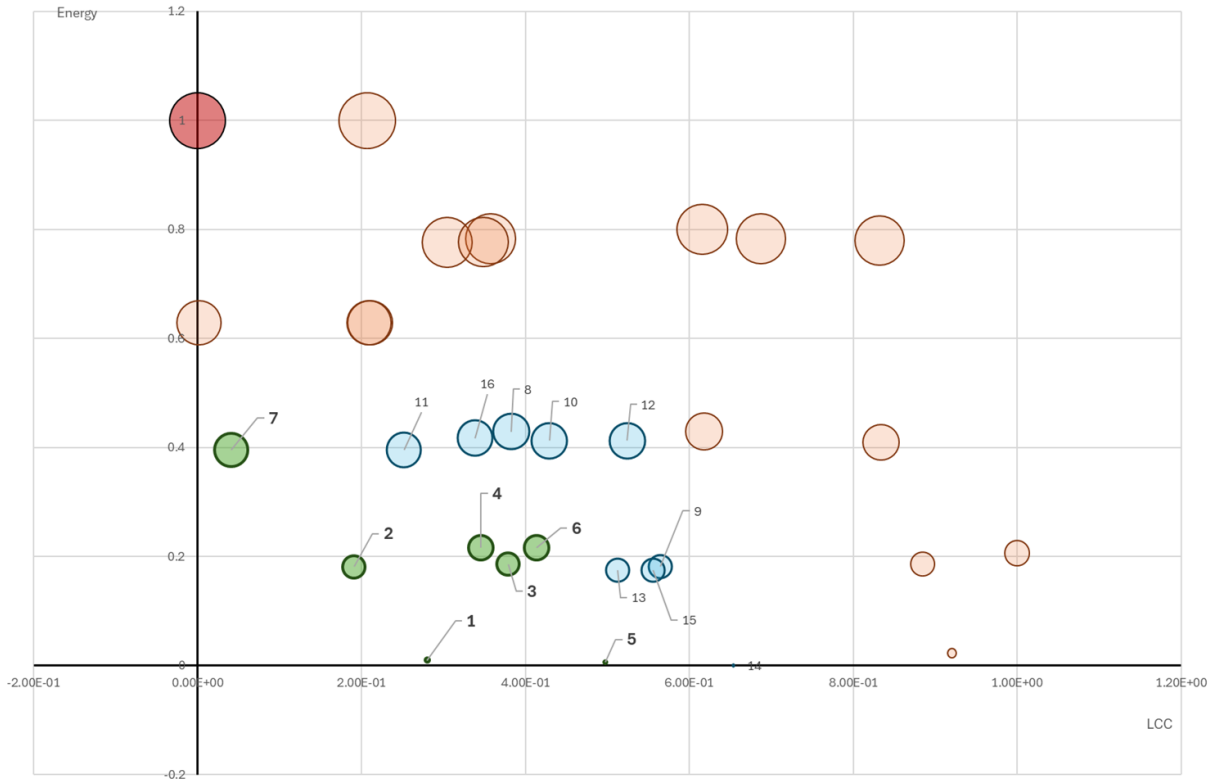
Rank	Window Type	Insulation Type	Insulation Side	Insulation Thickness	PV Peak	Energy saving	Extra Cost
1	Existing	No Change	No Change	No Change	16.9K Wp	35%	3%
2	Existing	GMW	Exterior	100mm	16.9K Wp	42%	9%
3	Double	No Change	No Change	No Change	16.9K Wp	35%	8%
4	Existing	GMW	Exterior	100mm	11.1K Wp	33%	7%
5	Triple	No Change	No Change	No Change	16.9K Wp	35%	10%
6	Existing	No Change	No Change	No Change	11.1K Wp	26%	1%



RESULTS - Apartment



Comparison: Normalised Energy-LCC-LCA for Apartment



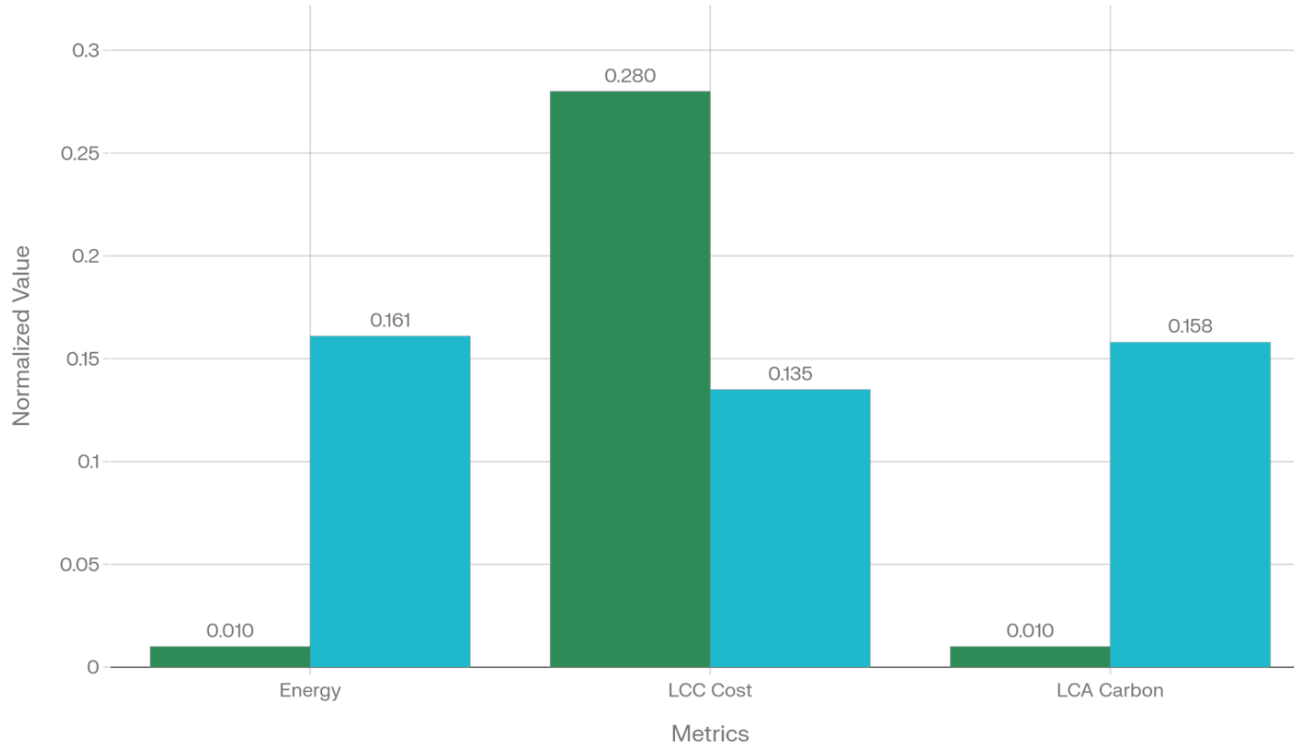
- Base case)
- Good
- Poor
- Excellent

sweet spot (low cost, low energy, low carbon)

Rank	Window Type	Insulation Type	Insulation Side	Insulation Thickness	PV Peak	Energy saving	Extra Cost
1	Existing	GMW	Interior	100mm	16.9K Wp	41%	6%
2	Existing	GMW	Interior	100mm	11.1K Wp	33%	4%
3	Existing	Hemp	Exterior	100mm	11.1K Wp	33%	8%
4	Double	No Change	No Change	No Change	16.9K Wp	32%	7%
5	Double	GMW	Exterior	100mm	16.9K Wp	41%	10%
6	Triple	No Change	No Change	No Change	16.9K Wp	32%	9%
7	Existing	No Change	No Change	No Change	11.1K Wp	25%	1%



CONCLUSION



Solar only (16 kWp PV) – no envelope changes needed



Insulation (GMW, 100mm interior) + 16 kWp PV

Best Design Findings:

- ✓ Apartment achieves superior performance through envelope investment (insulation) + PV
- ✓ Office optimal strategy is simpler: just add large solar array (cheaper retrofit, still ~35% energy savings)
- ✓ Trade-off difference: Apartment pays 2× more for upgrades but gets 16× better energy/carbon outcomes





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HÖGSKOLA



Tack så Mycket

(Thanks So Much)



MERRY CHRISTMAS!!

GOD JUL!!